

Testing and Validating Criteria and Results

Criteria for Testing	Feedback Received	Changes Made
Clarity and understanding of questions	Pilot participants (n=5) found Q12 "What would stop you?" had overlapping options. Two women said "Cost and husband decision are connected – husband decides because of cost."	Split Q12 into checkboxes with separate options. Added "Lack of trust in device" as a distinct option per pilot feedback. Reworded "Fear of technology" to "I am not comfortable using technology" for clarity.
Relevance to pregnant women in Kigali	Clinic manager noted Q8 "distance to health centre (km)" did not reflect reality – women measure travel time by moto or walking, not kilometres.	Added travel time option: "Less than 15 min / 15–30 min / More than 30 min" alongside km option. Both retained for literature comparability.
Sensitivity / distress potential	One pilot participant (4 months pregnant) said Q13 "Do you think a monitoring band could prevent pregnancy complications?" felt like a promise the device would work. She said, "What if it doesn't prevent and something happens?"	Changed wording to: "Do you think a monitoring band could help prevent complications?" Added "could help" to reduce overpromise. Added distress protocol reminder in researcher instructions.
Web app usability (Instrument 2)	Research assistants (n=2) reported that the Vercel app did not have a clear "save" confirmation after each entry. One assistant accidentally submitted incomplete data for two dummy participants.	Added a green confirmation message ("Response saved successfully") and a required "Confirm before submit" checkbox. Added auto-save every 30 seconds.

Consent clarity (both instruments)	Two pilot participants did not understand "withdrawal after submission is not possible." One asked, "What if I change my mind tomorrow?"	Added a simpler sentence to the consent introduction: "Once you press submit, I cannot delete your answers because no names are attached. Please take your time before submitting."
Mobile phone accessibility	Three pilot participants had basic phones (not smartphones) and could not access the Google Form link.	Instrument 2 (Vercel web app) designated as primary mode for non-smartphone users. Researcher enters responses on behalf of the participant using a tablet provided at the clinic.
